Artificial Intelligence

Assignment 5

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1 Prolog

Here are four different ways to extend our family example with child01/2 to arbitrary descendants:

1. descendantOf(X,Y):- childOf(X,Y).
   descendantOf(X,Y):- childOf(X,Z), descendantOf(Z,Y).

2. descendantOf(X,Y) :- descendantOf(Z,Y), childOf(X,Z).
   descendantOf(X,Y) :- childOf(X,Y) .

3. descendantOf(X,Y):- childOf(X,Y).
   descendantOf(X,Y) :- descendantOf(Z,Y), childOf(X, Z).

4. descendantOf(X,Y):- childOf(X,Z),descendantOf(Z,Y).
   descendantOf(X,Y) :- childOf(X,Y).

All those definitions are logically equivalent. Which ones work well in Prolog and which ones do not? Why?

The Prolog command trace. might help you

The 2nd definition leads to infinite recursion in the prolog.
The reason is first clause is recursive without a base case which results the program to keep searching for further descendants infinitely.
The below screenshot is the error shown in prolog.

```
Stack limit (0.2Gb) exceeded
  Stack sizes: local: 0.2Gb, global: 11.3Mb, trail: 0Kb
  Stack depth: 1,486,547, last-call: 0%, Choice points: 1,486,530
  In:
    [1,486,547] descendantOf(_1442, _1444)
    [1,486,546] descendantOf(_1468, _1470)
    [1,486,545] descendantOf(_1494, _1496)
    [1,486,544] descendantOf(_1520, _1522)
    [1,486,543] descendantOf(_1546, _1548)

Use the --stack_limit=size[KMG] command line option or
?- set_prolog_flag(stack_limit, 2_147_483_648). to double the limit.
```

Remaining definitions works well and are logically equivalent.

2 Default rules (6 Points)

Represent the following sentences as default rules:

1. Spouses of Germans rarely hate soccer.
2. People usually have the same hometown as their spouse.
3. Most of the time, a person's hometown is where his or her employer is located.

Use predicate symbols spouse/2, employer/2, german/1 and hate/2, as well as function symbols hometown/1, location/1 and soccer/0.

1. Spouses of Germans rarely hate soccer.

$$\frac{\text{spouse}(X,Y) \land \text{german}(Y) : \sim\text{hate}(X,\text{soccer})}{\sim\text{hate}(X, \text{soccer})}$$

2. People usually have the same hometown as their spouse.

$$\frac{\text{spouse}(X,Y) : \text{hometown}(X) = \text{hometown}(Y)}{\text{hometown}(X) = \text{hometown}(Y)}$$

3. Most of the time, a person's hometown is where his or her employer is located.

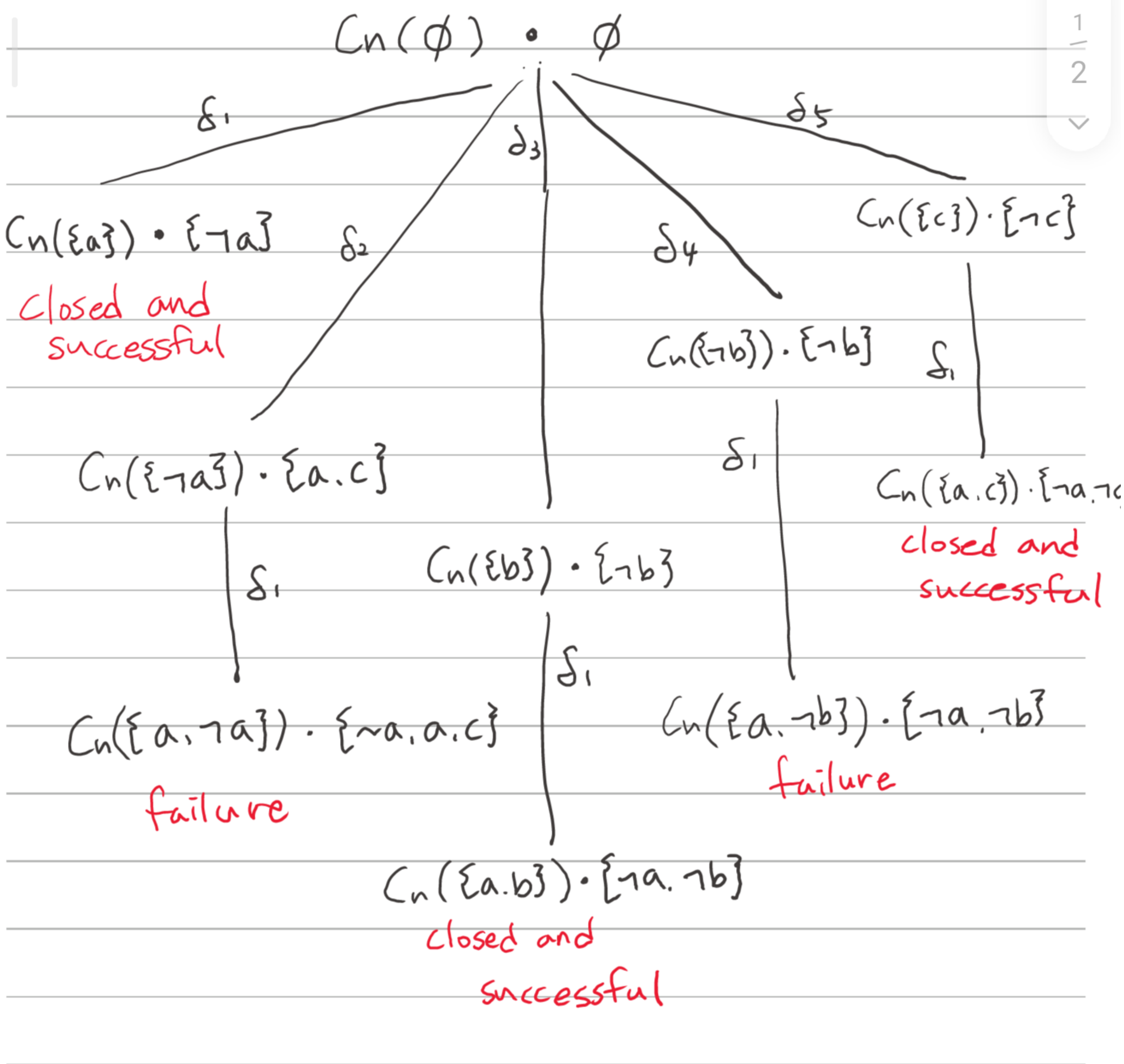
$$\frac{\text{employer}(X,Y) : \text{hometown}(Y) = \text{location}(X)}{\text{hometown}(X) = \text{location}(X)}$$

3 Default theory (10 Points)

Let $T = (W, \Delta)$ be a default theory with $W = \emptyset$ and $\Delta = \{\delta_1, \delta_2, \delta_3, \delta_4, \delta_5\}$ with

$$\delta_1 = \frac{\top \ : \ a}{a}, \quad \delta_2 = \frac{\top \ : \ \neg a, \ \neg c}{\neg a}, \quad \delta_3 = \frac{a \ : b}{b}, \quad \delta_4 = \frac{a \ : \ \neg b}{\neg b}, \quad \delta_5 = \frac{\neg a \ : \ c}{c}$$

Calculate all extensions of T using a process tree. Make sure that you both draw the tree and state the resulting extensions.



Extension 1: {a, b}
Extension 2: {a, c}